

Insert Study-Specific Title Here

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Abstract—

Index Terms—medical AI, laboratory medicine, machine learning, clinical prediction, uncertainty, explainable AI

I. INTRODUCTION

- A. *Clinical/scientific background*
- B. *Problem statement*
- C. *Existing evidence*
- D. *Research gap*
- E. *Study objective*
- F. *Hypothesis*
- G. *Contributions*

II. RELATED WORK

- A. *Clinical prediction and medical AI*
- B. *Laboratory-based modeling*
- C. *Missing-data and missingness-aware methods*
- D. *Uncertainty and calibration*
- E. *Explainability*
- F. *Adaptive evidence acquisition*
- G. *Position of the present study*

III. MATERIALS AND METHODS

- A. *Study design*
- B. *Research setting and data sources*
- C. *Dataset description*
- D. *Ethics and data governance*
- E. *Population and cohort construction*
- F. *Inclusion and exclusion criteria*
- G. *Index date and observation window*
- H. *Outcome definition*
- I. *Predictor variables*
- J. *Laboratory representation*
- K. *Reference ranges and units*
- L. *Temporal feature construction*
- M. *Missingness representation*
- N. *Data preprocessing*
- O. *Data leakage prevention*
- P. *Train/validation/test strategy*
- Q. *External validation*

IV. MODEL DEVELOPMENT

- A. *Reference/rule-based baseline*
- B. *Statistical baselines*
- C. *Machine learning models*
- D. *Deep learning models*
- E. *Ensemble models*
- F. *Hyperparameter optimization*
- G. *Class imbalance handling*
- H. *Calibration methods*
- I. *Uncertainty quantification*
- J. *Ablation studies*

V. ADAPTIVE EVIDENCE ACQUISITION METHODOLOGY

- A. *Problem formulation*
- B. *Candidate evidence set*
- C. *Evidence-value metric*
- D. *Sequential decision policy*
- E. *Stopping criterion*

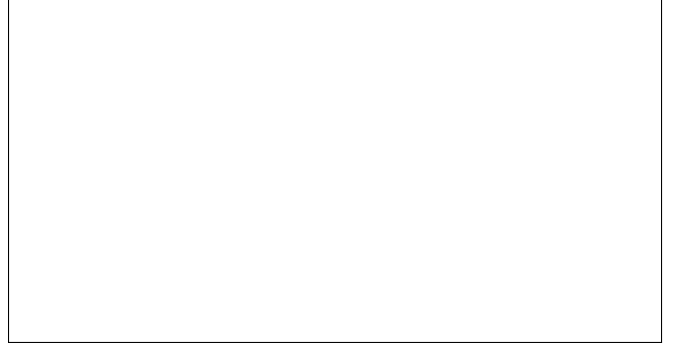


Fig. 1. Placeholder for the principal model/analysis figure.

Validation; Visualization; Writing—Original Draft; Writing—Review & Editing.

Sababa Ateeq: Conceptualization; Methodology; Biomedical Interpretation; Investigation; Validation; Writing—Original Draft; Writing—Review & Editing.

ACKNOWLEDGMENT

APPENDIX A

SUPPLEMENTARY METHODS

- A. *Preprocessing details*
- B. *Hyperparameter search*
- C. *Additional ablations*

APPENDIX B

ADDITIONAL RESULTS

TABLE I
PRIMARY MODEL PERFORMANCE. REPLACE PLACEHOLDER VALUES WITH ESTIMATES AND CONFIDENCE INTERVALS.

Model	AUROC	AUPRC	Sensitivity	Specificity	Brier	Calibration slope
Baseline	–	–	–	–	–	–
Logistic/Elastic Net	–	–	–	–	–	–
Tree model	–	–	–	–	–	–
Deep model	–	–	–	–	–	–
Proposed model	–	–	–	–	–	–